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MOLECULAR DESIGN OF A HIGHLY STABLE SINGLE-ION CONDUCTING POLYMER GEL ELECTROLYTE

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ABSTRACT

Single-ion conducting (SIC) polymer electrolytes with a high Li transference number (t_{Li^+}) have shown the capability to enable enhanced battery performance and safety by avoiding liquid-electrolyte leakage and suppressing Li dendrite formation. However, issues of insufficient ionic conductivity, low electrochemical stability, and poor polymer/electrode interfacial contact have greatly hindered their commercial use. Here, a Li-containing boron-centered fluorinated SIC polymer gel electrolyte (LiBFSIE) was rationally designed to achieve a high t_{Li^+} and high electrochemical stability. Owing to the low dissociation energy of the boron-centered anion and Li^+ , the as-prepared LiBFSIE exhibited an ionic conductivity of 2×10^{-4} S/cm at 35°C, which is exclusively contributed by Li ions owing to the high t_{Li^+} of 0.93. Both simulation and experimental approaches were applied to investigate the ion diffusion and concentration gradient in the LiBFSIE and non-cross-linked dual-ion systems. Typical rectangular Li stripping/plating voltage profiles demonstrated the uniform Li deposition assisted by LiBFSIE. The interfacial contact and electrolyte infiltration were further optimized with an *in situ* UV-vis-initiated polymerization method together with the electrode materials. By virtue of the high electrochemical stability of LiBFSIE, the cells achieved a promising average coulombic efficiency of 99.95% over 200 cycles, which is higher than that of liquid-electrolyte-based cells. No obvious capacity fading was observed, indicating the long-term stability of LiBFSIE for lithium metal batteries.

INTRODUCTION

Because of their high energy densities, lithium-ion batteries have become the ubiquitous energy storage form for electric vehicles and personal portable electronics. The state-of-the-art choice of cathodes is Li intercalation transition metal oxide materials, such as lithium iron phosphate (LFP), $\text{LiNi}_{0.8}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2$ (NCA), $\text{LiNi}_x\text{Mn}_y\text{Co}_z\text{O}_2$ ($x+y+z=1$, NMC), and LiCoO_2 (LCO) and so on.¹

To push the energy-density limit of current lithium-ion batteries, a common practice is to increase the working voltage. Another approach, however, is to use lithium metal as the anode; with a theoretical gravimetric specific capacity of 3860 mAh/g, lithium metal is considered the “Holy Grail” in the lithium-ion battery community.² There are many issues with using lithium metal as the anode, such as dendrite formation, rapid impedance buildup, unstable solid electrolyte interphase (SEI) leading to consistent consumption of electrolytes, and high safety hazard.^{3,4} In this context, an electrolyte with high electrochemical stability and safety is urgently required.^{5,6} Solid-state electrolytes (SSE), with their all-solid and Li-ion-conducting nature, have been regarded as a potential candidate for next-generation batteries. After lithium phosphorus oxynitride (LiPON) was synthesized in the 1990s,^{7,8} a series of ceramic electrolytes were developed including perovskite,⁹ sodium superionic conductor (NASICON),^{10,11} sulfide,^{12,13} garnet,^{14,15} and others. Considering the manufacturing convenience for cell integration and the cost, polymer electrolyte is regarded as one of the most promising approaches.¹⁶ Since ionic transport behavior was observed in a “plastic” solid material, poly(ethylene oxide) (PEO), by Wright et al. in 1973¹⁷ and such a material was utilized as an electrolyte for batteries by Armand et al. in 1978,¹⁸ a number of efforts have been applied such as investigation of Li solvation by oxyethylene,¹⁹ studies of the salt effect in forming complexes,²⁰ and exploration of other polymer systems including poly(methyl methacrylate) (PMMA),^{21,22} poly(acrylonitrile) (PAN),^{23,24} and poly(vinylidene fluoride) (PVDF).^{25,26} However, a low lithium transference number (t_{Li^+}) and low ion diffusivity result in steep concentration gradients, leading to undesired polarization and side reactions.²⁷ Another issue is the compatibility of the polymer gel electrolyte with the Li metal anode: It is known that at least twice the shear modulus of the Li metal electrode is usually required for a solid electrolyte to

suppress formation of Li dendrites,²⁸ which are a challenge for polymer electrolyte design because of their intrinsically low mechanical strength.

Single-ion conducting (SIC) polymer electrolyte, with its high t_{Li^+} , has been recognized for its potential application to state-of-the-art Li-ion batteries.²⁹ Because of its high t_{Li^+} , a homogeneous ion concentration distribution can be achieved to guide uniform Li deposition. It was shown that Li dendrite growth could be eliminated with a SIC electrolyte of low mechanical strength.³⁰ Bis(trifluoromethyl sulfonyl)imide(TFSI⁻)-based polymers have been intensively studied because they display weaker coordination between Li⁺ and the anions^{31–35} compared with other systems, such as those with carboxylate anions^{36–38} and sulfonate anions.^{39–41} Bouchet et al. reported the use of a polystyrene (PS)-PEO block copolymer with TFSI⁻ bonded to PS, where the microphase-separated PEO molecules are employed for conducting Li⁺.⁴² A t_{Li^+} above 0.85 was achieved with an ionic conductivity of 1.3×10^{-5} S cm⁻¹ at 60°C. Iojoiu and his coworkers explored a multi-block copolymer single-ion conductor with aromatic fluorinated structures.⁴³ By using a significant amount of ethylene carbonate (EC, 65 wt%), they obtained a remarkable ionic conductivity of 10^{-3} S/cm at RT, and a Li/NMC cell equipped with the multi-block copolymer lasted for 230 cycles, with a coulombic efficiency subsequently reaching 99.5%. However, the issue of Al corrosion with TFSI⁻ cannot be neglected.⁴⁴ SIC polymer electrolytes with boron- and phosphorus-based anions have relatively low dissociation energy, with the potential to promote Li conduction within the polymer host.⁴⁵ Qin et al. have synthesized EC/dimethyl carbonate (DMC) swollen polymeric lithium pentaerythrite borate @ poly(vinylidene fluoride-co-hexafluoropropene) (PVDF-HFP) and polymeric lithium di(trimethylolpropane)borate (PLDB) @ PVDF-HFP membranes that exhibited a high t_{Li^+} of 0.89 and the ability to passivate Al.⁴⁶ Cheng's group reported studies of a lithium poly(bisphenol A borate) (LiBPAB)/PEO polymer blend electrolyte; the interplay between LiBPAB and PEO allows the Li to hop through the ethylene oxide segments or transport with the

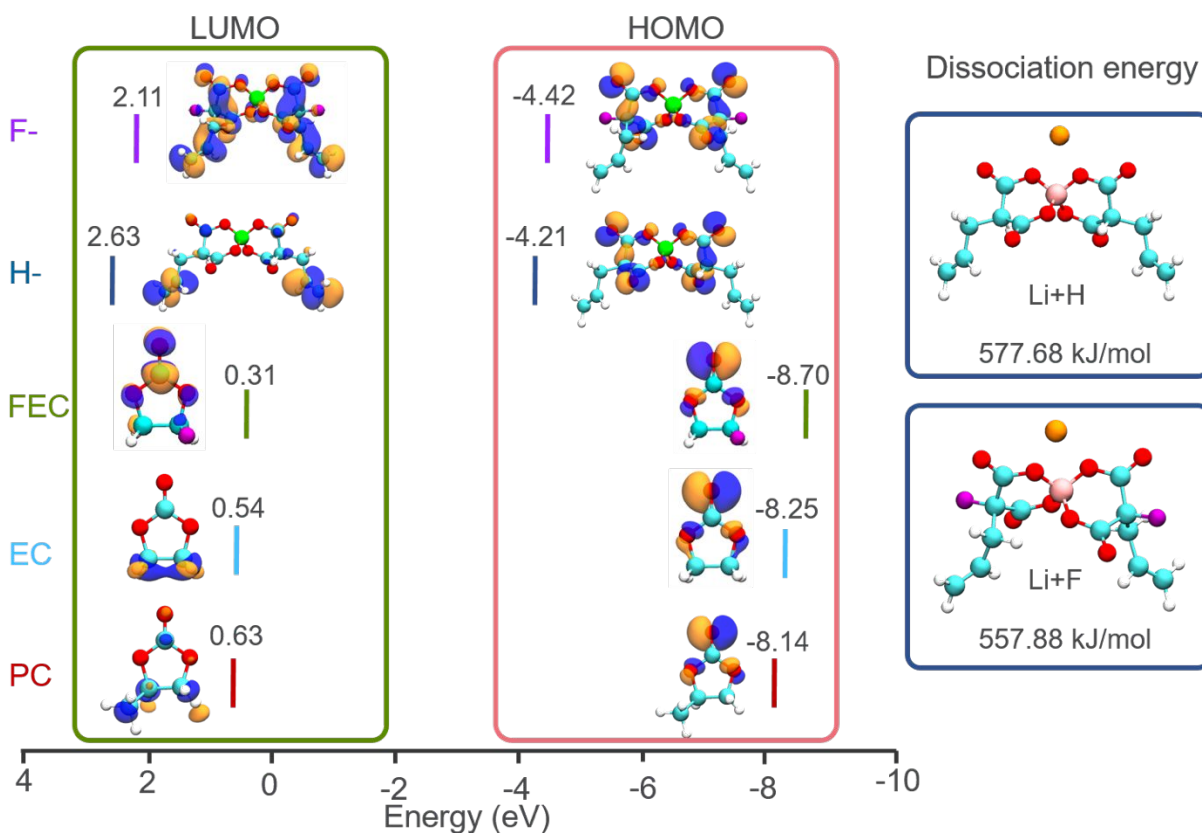
boron atoms, assisted by ethylene oxide segments.⁴⁷ Recently, Ignacio et al. synthesized a SIC electrolyte based on sp^3 boron and poly(ethylene glycol) (PEG) via condensation reactions.⁴⁸ An increase in t_{Li^+} and electrochemical stability was observed as the boron proportion increased and the PEG length decreased. However, syntheses of boron-based polymers are usually tedious and require high temperature and further purification processes. On the other hand, composite SIC electrolytes were also explored with the aim of improving the mechanical strength of the electrolyte.^{49,50} Despite considerable efforts in the past few years to develop a polymer structure to immobilize the anions, there are still issues from the perspective of electrolyte design or processing, such as electrolyte/electrode interfacial resistance, electrochemical stability of both the cathode and the low-redox-potential Li metal anode, the feasibility of the processing method with respect to current battery manufacturing techniques, and so on. The present study has tried to solve these problems through (1) molecular engineering to design the basic unit of the polymer framework, (2) cell fabrication with the goal of alleviating the interfacial resistance, and (3) electrochemical testing. An efficient and defect-free polymerization method has been developed to achieve a high-quality Li-containing boron-centered fluorinated SIC polymer gel electrolyte (LiBFSIE). *In situ* integration of LiBFSIE with electrodes was achieved, enabling excellent wetting and infiltration to solve the contact issues at the interphase. Computational and electrochemical characterization methods were applied to evaluate the functionality of the LiBFSIE in a battery. A promising average coulombic efficiency of 99.95% was achieved in a Li/LFP cell.

RESULTS AND DISCUSSION

Rational Design for SIC Polymer Gel Electrolyte

To obtain high-quality polymeric LiBFSIE material, thiol-ene free-radical polymerization was adopted as an efficient, defect-free, and high-yield polymerization method, in which an olefin undergoes anti-Markovnikov addition of an alkyl thiol (RS-H).⁵¹ The key is to use multifunctional thiol end molecules together with α,ω -dienes. Therefore, lithium bis(fluoroallyl)malonato borate (LiBFMB) with α,ω -dienes, and trimethylolpropane tris(3-mercapto propionate) (TMPT) with three active thiol ends, were selected for the UV-initiated polymerization (**Figure S1 in Supporting Information**). In the design of the functional and polymerizable Li salt, a non-coordinating sp^3 boron center of LiBFMB was synthesized to achieve high ionic conductivity due to the low dissociation energy between boron and lithium. The introduction of a fluorinated structure is intended to increase the anodic stability of the Li salt monomer, owing to the strong electron-withdrawing nature of fluorine. A mixture of propylene carbonate (PC)/ ethylene carbonate (EC)/ fluoroethylene carbonate (FEC) was employed to dissolve the salt as precursors and behave as plasticizers in the polymer electrolyte. A computational approach was applied to determine the highest occupied molecular orbital (HOMO) and lowest unoccupied molecular orbital (LUMO) of BFMB⁻, bis(allyl)malonato borate (BMB⁻), and plasticizers from the optimized geometry (**Scheme 1**). (See details in experimental section.) As expected, a lower HOMO energy level was observed when introducing a high-energy-withdrawing group, fluorine, into the anion monomer, indicating the oxidative stability of BFMB⁻ at high voltage. Moreover, the chemical stability of the Li-salt monomer is also improved as a result of lowering the acidity via the substitution of a proton with fluorine. The optimized geometries of LiBMB and LiBFMB are illustrated in **Figure S2**. Meanwhile, the high ionic conductivity of the polymer gel electrolyte can

be expected because of the lower dissociation energy between Li^+ and BFMB^- (i.e., 557.8 kJ/mol) compared to that between Li^+ and BMB^- (i.e., 577.7 kJ/mol). The comparisons of the dissociation energies of various ion pairs are listed in **Table S1**. This result also reconciles well with the potential of mean force (PMF) profile as a function of the distance between the Li^+ ion and the boron atom in the BFMB ion, using classical molecular dynamics simulation. The energy barrier (marked in **Figure S3b**) for the escape of the Li^+ ion from the center boron is only ~ 8 kJ/mol.



Scheme 1. Diagram of the HOMO and LUMO energy levels, showing the electron acceptance and donation abilities of PC, EC, FEC, BMB^- and BFMB^- . A lower HOMO energy level is demonstrated in BFMB^- compared with BMB^- , indicating the oxidative stability at high voltage.

***In situ* Preparation of LiBFSIE/Electrode**

A high-yield and cost-effective method was developed to synthesize LiBFMB. (See **Figure S4**; details are described in the experimental section.) The chemical structure of LiBFMB is examined by ^1H , ^{11}B , and ^{19}F NMR spectra in **Figure S5**. The cross-linked polymer gel electrolyte was prepared via a “click” reaction between TMPT (the cross-linker) and the alkene groups of the Li salt monomer. (Details are described in the experimental section.) UV-initiation was applied to generate the thiyl radical that propagated with the -ene group. Owing to the high efficiency and yield of thiol-ene reactions, Li salt monomers (LiBFMB) were all anchored into the network and lost their mobility, hence the name single-ion network. Meanwhile, the plasticizer molecules trapped in the polymer framework could further improve Li transport.

The contact between the electrolyte and electrode materials is one critical issue that hinders the application of solid electrolytes (either ceramic- or polymer-based). To ensure good contact between the electrolyte and electrode surface, an *in situ* processing method was explored: the SIC polymer precursor, which has a low viscosity, was drop-casted onto the Li metal and the LFP cathode separately, as shown in **Figure 1a**. This process allows the liquid polymer precursor to accommodate the porosity and rough surface of the composite cathode disk through the wetting and infiltration processes. After the UV-vis cross-linking, the LiBFSIE-Li and LiBFSIE-LFP electrodes (the insets in **Figure 1b** and **1c**) were compressed together and fabricated into a coin cell. The interface between polymer gel electrolyte and electrode surface was characterized by scanning electron microscopy (SEM) and marked with a dotted line (**Figure 1b** and **1c**). From the cross-section view, the thickness of the polymer gel electrolyte on the Li metal and LFP electrode was determined to be 130 μm and 40 μm , respectively. The thickness of the LiBFSIE layer can be adjusted by controlling the amount of the precursor when casting. No obvious gap was witnessed

on either the bare Li metal surface or the porous LFP cathode surface, suggesting good contact at the interface. Also, when integrating the anode and cathode, the LiBFSIE was found to have an adhesive nature, enabling good contact at the interface between the LiBFSIE layers. To further characterize the mechanical properties and binding behavior of LiBFSIE, free-standing LiBFSIE film was also prepared following the same protocol via UV-cross-linking reactions in Teflon molds with custom-designed geometries (**Figure 2a and 2b**). The mechanical strength of the LiBFSIE polymer gel electrolyte was tested by stretching the film at a constant elongation rate of 5 mm/min with an Instron tensile instrument as shown in **Figure S6**. The stress exhibited a linear relationship with the increase in the strain before cracking, and the polymer gel electrolyte exhibited a Young's modulus of 383 kPa. Furthermore, two pieces of isolated polymer gel electrolyte films were partially attached and placed on the Instron instrument for a typical stretching test (**Figure S6a**). A similar modulus of the single piece (**Figure S6b**) was observed, followed by partial and full detachment of these two pieces, indicating the good adhesiveness of the polymer/polymer gel electrolyte. (The detailed calculation of the modulus is discussed in the Supplementary Information.) In contrast, such an adhesive nature was not observed when LIBFSIE film was directly applied to the Li metal or LFP electrode.

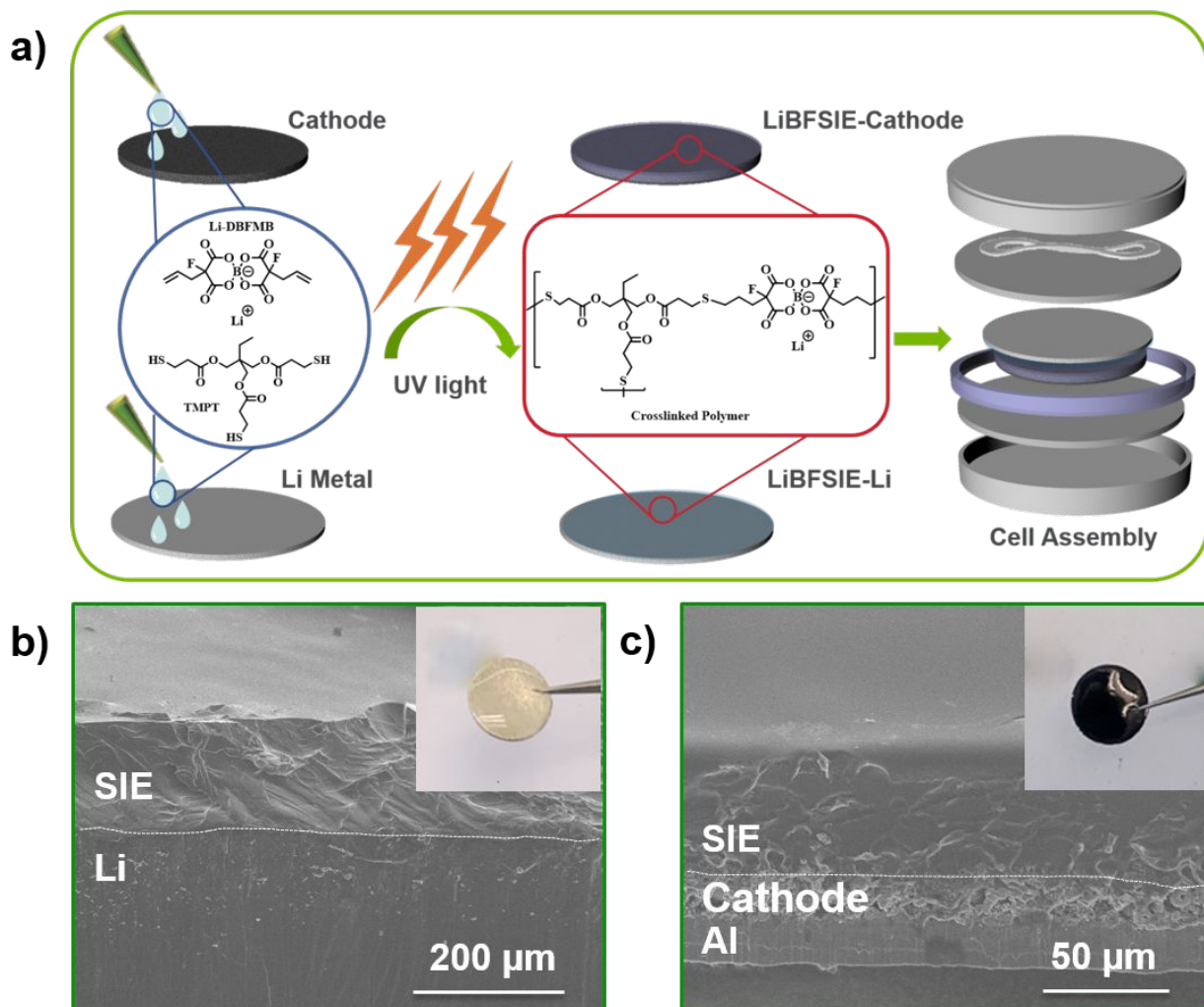


Figure 1. (a) Schematic illustration of the *in situ* coating of the LFP cathode and Li metal electrode with LiBFSIE, and cell assembly. The polymer precursors were drop-casted onto the cathode and Li metal electrode surfaces before the UV exposure, enabling wetting of the electrodes. After the cross-linking reaction, the LiBFSIE-coated cathode and anode were compressed together and assembled into a coin cell. SEM images of the cross-sections of (b) LiBFSIE-Li and (c) LiBFSIE-cathode show the good contact at the electrode-electrolyte interface. The insets are optical images of LiBFSIE coatings on Li metal and the cathode, respectively. The shiny surface comes from the LiBFSIE electrolyte, and is also observed on the free-standing LiBFSIE in **Figure 2a** and **2b**.

Fourier-transform infrared (FTIR) spectroscopy was employed to further examine thiol-ene polymerization (**Figure 2c** and **Figure S7**). The peaks at 3080 cm^{-1} (c1) and 1640 cm^{-1} (c2) are attributed to the -C-H stretching and -C=C- stretching from LiBFMB. While the -S-H stretching from TMPT was observed originally at 2570 cm^{-1} (c3), these peaks disappeared after the cross-linking reaction because of the consumption of the allyl and thiol groups, indicating a nearly full conversion.

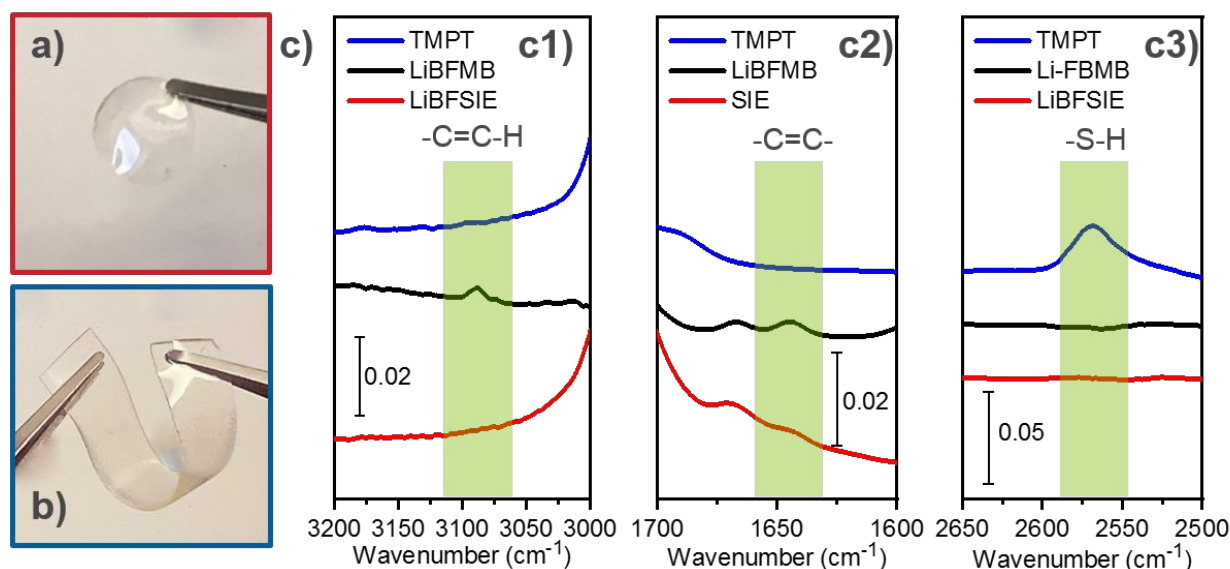


Figure 2. Chemical structure analysis of LiBFSIE. Optical images of flexible LiBFSIE prepared from round (a) and rectangular (b) molds. (c) FTIR spectrum comparisons of TMPT (blue curve), LiBFMB (black curve), and LiBFSIE (red curve). The absence of alkene -C-H stretching at 3080 cm^{-1} (c1) and -C=C- stretching at 1640 cm^{-1} (c2) from LiBFMB, and the -S-H stretching at 2570 cm^{-1} (c3) from TMPT, indicate the near-completion of the thiol-ene reaction.

Electrochemical Properties

The ionic conductivity of the freestanding LiBFSIE film was measured via impedance spectroscopy in a symmetric setup of stainless steel (SS)/LiBFSIE/SS at indicated environmental

temperatures. A promising ionic conductivity of 2.0×10^{-4} S/cm is achieved at 35°C. The linear relationship between the measured ionic conductivity and the temperature demonstrated a typical Arrhenius behavior, and the low activation energy was determined to be 11.09 kJ/mol (**Figure 3a**), which is consistent with the number independently obtained from X-ray photon correlation spectroscopy (XPCS) (**Figure S8**). The relatively low activation energy indicates fast ionic transport along the conductive channels inside the LiBFSIE polymer gel electrolyte,⁵² which could be related to the low dissociation energy of the sp^3 boron-centered anion with Li^+ and enhanced chain/molecular movement by plasticizers. A comparison of the activation energy of LiBFSIE with that of previously reported polymer gel electrolytes is demonstrated in **Table S2**. A temperature of 35°C was employed for the following electrochemical property characterization and battery tests, considering that the optimum handling temperature for electric devices is usually between 0°C and 35°C. The electrochemical stability of the LiBFSIE was characterized by a cyclic voltammogram at a scan rate of 0.2 mV/s from -0.5 V to 5 V for three cycles (**Figure 3b**). The cathodic peak at -0.5 V and anodic peak at 0.5 V are attributed to the Li plating and stripping. No obvious oxidation peaks were witnessed, even at 4.8 V (inset in **Figure 3b**), indicating the excellent electrochemical stability of LiBFSIE. Note that a planar SS electrode was applied as the working electrode; therefore, the stability could be overestimated, considering the heterogeneous porous structure of the cathode and its exacerbated reactivity after delithiation.^{53,54} The Li transport phenomenon inside the polymer gel electrolyte usually depends on both the salt concentration and ionic conductivity. In our case, a cross-linked single-ion network is present while Li^+ is anchored on the delocalized borate anion center, and no extra Li^+ resource is required.

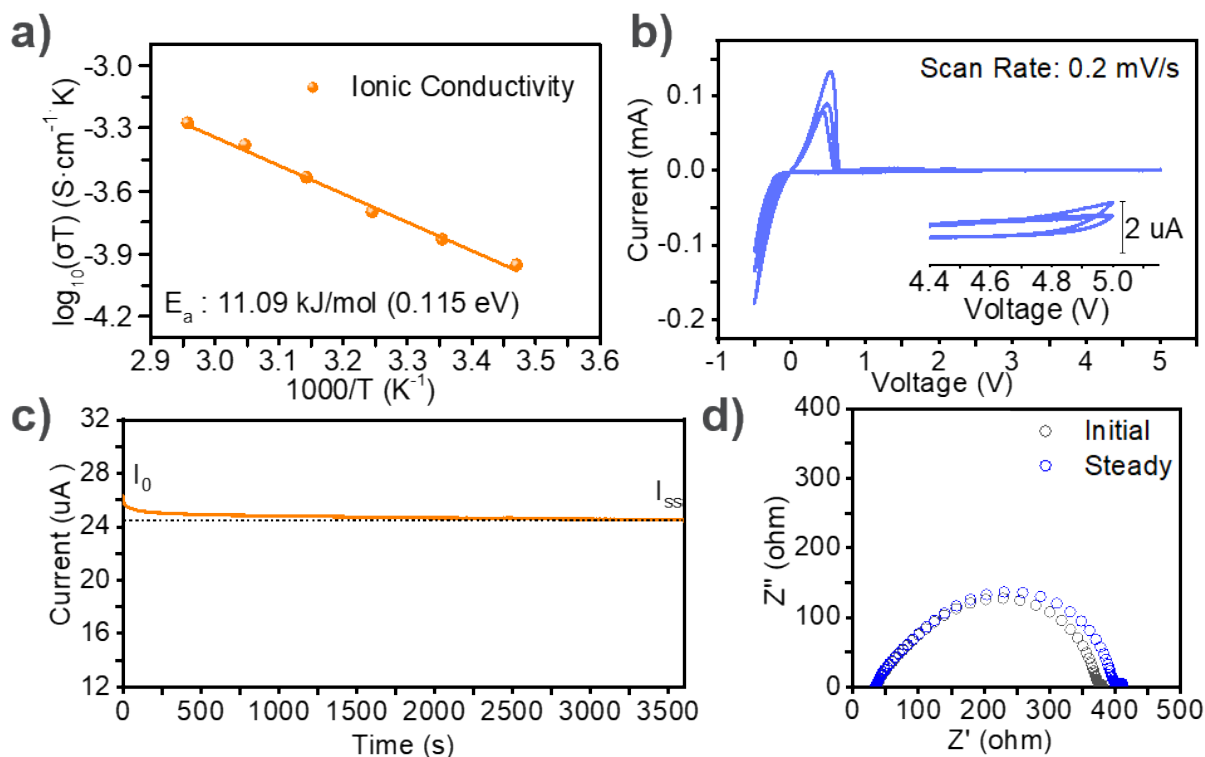


Figure 3. Analysis of the electrochemical properties of LiBFSIE. (a) Ionic conductivities of LiBFSIE with varying temperatures from 15°C to 65°C. The activation energy was determined to be 11.09 kJ/mol. (b) Cyclic voltammogram of a Li/LiBFSIE/SS asymmetric cell in a chemical window from -0.5 V to 5 V at a scan rate of 0.2 mV/s. c) Chronoamperogram of a Li/LiBFSIE/Li symmetric cell with an applied voltage of 10 mV. I_0 and I_{ss} indicate the initial current and the steady-state current. d) Nyquist plot of Li/LiBFSIE/Li symmetric cell before/after polarization. R_0 and R_s refer to the initial and steady-state internal resistance.

Based on our molecular design, the transference would be governed entirely by the cation species, Li⁺. The t_{Li^+} of the polymer gel electrolyte was measured via a potentiostatic polarization method, in which the current and internal resistance at initial and steady-state were recorded when a small constant potential was applied (**Figure 3c and 3d**). Using Bruce and Vincent's equation,⁵⁵ the t_{Li^+}

of LiBFSIE was calculated to be 0.93, which is significantly higher than for the non-cross-linked electrolyte, with a t_{Li^+} around 0.48 (**Figure S9**). (Non-cross-linked electrolyte was prepared with the same recipe as LiBFSIE but without UV cross-linking). The SIC behavior of the LiBFSIE was further demonstrated via a galvanostatic DC polarization test. Theoretically, in the SIC system, there is an exact square voltage response to a DC current, assuming that t_{Li^+} is equal to 1 when there is no ion diffusion (the ion movement is driven by the concentration difference), and since all of the voltage rise is contributed by the migration (the ion movement in the presence of an electric field). In the case of conventional binary-ion systems, the voltage jump is accompanied by a gradual slope because of the ion diffusion while applying a direct current. For example, the ion diffusion driven by the electric fields results in the ion concentration gradient and induces an impedance change which requires additional potential input to maintain constant currents.⁵⁶ In this work, voltage profiles were recorded in symmetric Li cells for LiBFSIE and non-cross-linked systems at constant indicated currents (**Figure 4a and 4b**). (Small current densities and short static times were applied to minimize the effect of the morphology change of Li metal.) For LiBFSIE, stepwise voltage profiles with slight potential increases were observed, in which the voltages rose from initial values of 5.74 mV, 22.67 mV, and 45.07 mV and equilibrated at 5.93 mV, 23.39 mV and 46.24 mV under current densities of 0.01, 0.04 and 0.08 mA/cm², respectively. In comparison, obvious gradual voltage rises of non-cross-linked liquid electrolytes were observed (**Figure 4b**). It is worth mentioning that the lower overpotential of the non-cross-linked electrolyte compared with the LiBFSIE could be associated with the motion of the anions, which results in an increase of the ionic conductivity but do not contribute to the Li transport (**Figure 4a1-3 and 4b1-3**). To clarify the ion diffusion and migration behavior of the electrolyte, voltage profiles are normalized for different current densities, and the proportions of the voltage increase contributed by migration

and diffusion are demonstrated in **Figure 4a1-3** and **4b1-3**. (The overpotentials induced by diffusion and migration are marked with yellow and blue, respectively.) For LiBFSIE, the potential increase induced by ion diffusion only accounted for 3.71% of the total voltage rise (0.01 mA/cm²), whereas this number in the non-cross-linked liquid electrolyte is 16.07% (0.01 mA/cm²). This larger proportion attributable to diffusion indicated that the severe concentration gradient resulted from the anion diffusion. To provide insights into concentration gradients induced by ion diffusion under an applied electric field, a finite-element method (COMSOL Multiphysics) was utilized to investigate the Li ion distribution. (Detailed simulation methods are discussed in the Supplementary Information.) Under ideal circumstances, when t_{Li^+} exactly equals 1, no gradients will appear, since there is not a single mobile anion. In our case, although the t_{Li^+} is 0.93, there are still a small number of electrons carried by anions during the potential static test. Therefore, a slight concentration variation was found after 120 s (**Figure 4c**). On the other hand, steep Li⁺ concentration gradients were observed in the system of non-cross-linked liquid electrolyte although its thickness is less than half of that in LiBFSIE: the concentration of Li⁺ became more and more polarized as the charging time increased (**Figure 4d**). The evolution of ion concentration gradients over the charging time of LiBFSIE and the non-cross-linked electrolyte is illustrated in Video S1 and S2, respectively. By comparing the experimental and simulation results on Li transport behaviors in LiBFSIE and a non-cross-linked liquid electrolyte system, it is seen that the anchoring of anions into the polymer network dramatically eliminated undesired ion diffusion and enabled a uniform Li⁺ concentration distribution in the cell. This uniformity was able to not only suppress the undesired side reactions but also guide uniform Li deposition, resulting in high stability and coulombic efficiency.

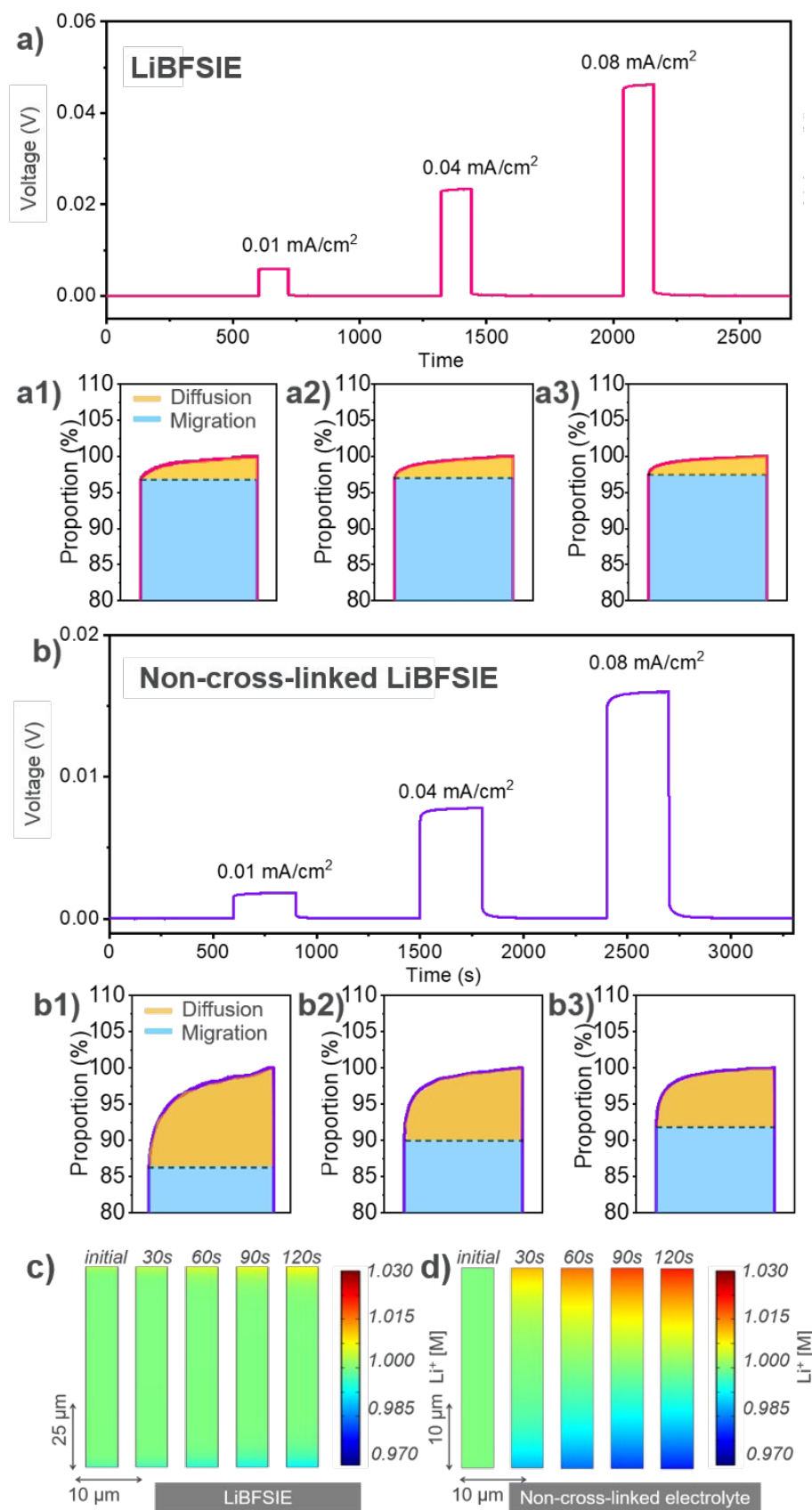


Figure 4. Investigation of Li transport and ion diffusion in LiBFSIE and non-cross-linked liquid electrolyte. a, b) Voltage profiles of Li/LiBFSIE/Li and Li/non-cross-linked liquid electrolyte/Li symmetric cells at indicated currents. Additional polarization time was employed for non-cross-linked liquid electrolyte because of the gradual voltage increase. a1-3, b1-3) Normalized voltage profiles for LiBFSIE and non-cross-linked liquid electrolyte. The overpotentials induced by diffusion and migration are shown in yellow and blue, respectively. c, d) Li^+ concentration distribution profiles of LiBFSIE and non-cross-linked liquid electrolyte in Li symmetric cells at 0.08 mA/cm^2 . The finite-element method is applied and conducted via COMSOL.

Li Stripping and Plating

Lithium plating/stripping measurements were performed to evaluate the compatibility of LiBFSIE with Li metal. It is well known that dendrites are formed on a bare Li metal electrode in a thermodynamically favored manner;⁵⁷ an unstable SEI layer on the Li anode also leads to the rapid loss and/or consumption of both active Li and electrolyte.⁵⁸ SICs, with their high t_{Li^+} , were expected to promote Li deposition and form a stable SEI. In this work, the symmetric Li-LiBFSIE/LiBFSIE-Li cells were used to plate and strip Li at the constant current densities indicated in **Figure 5a**, with 0.5 h for each stripping/plating. The detailed voltage profiles, with indicated current densities and cycle times, are shown in **Figure 5b-e**. The overpotential increases from 0.014 V to 0.095 V, 0.21 V, and 0.48 V along with the increase of current density from 0.02 mA/cm^2 to 0.1 mA/cm^2 , 0.2 mA/cm^2 , and 0.4 mA/cm^2 , respectively. Even though there is an uneven voltage at 0.4 mA/cm^2 which could be attributed to the ionic conductivity limit of the LiBFSIE, rectangle-like voltage profiles with an overpotential of 0.11 V were demonstrated when the current returned to 0.1 mA/cm^2 ; these profiles lasted for 300 cycles without any apparent

Battery Performance of Li/LiBFSIE/LiFePO₄

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Li/LiBFSIE/LiFePO₄ cell with a 200- μ m-thick LiBFSIE layer is illustrated in **Figure 6a**. The cell delivered a reversible capacity of 137.9 mAh/g at 0.1 C, 135.0 mAh/g at 0.2 C, 131.1 mAh/g at 0.4 C, and 124.7 mAh/g at 0.7 C. At a higher current density of 1.5 C, a capacity of 106.4 mAh/g was maintained. When the current density was set back to 0.4 C, the capacity recovered back to 131.6 mAh/g, revealing the high reversibility of the LFP cathode paired with the LiBFSIE polymer gel electrolyte. Moreover, when a lesser amount of the LiBFSIE precursor was employed, a thinner LiBFSIE coating was achieved, with a total thickness of around 100 μ m. Better rate capabilities were observed when the thickness of LiBFSIE was reduced. Therefore, a harsher battery testing condition was applied in **Figure 6b**: a remarkable reversible capacity of 96.5 mAh/g was achieved up to 2.5 C, returning to 124.6 mAh/g at 1 C. Charge/discharge profiles are illustrated in **Figure S10**. Long-cycling characterization of the cells was performed at 0.5 C followed by formation cycles at 0.15 C. The cell exhibited a reversible capacity of 125.4 mAh/g after 200 cycles without significant degradation (**Figure 6c**). The coulombic efficiency of the LFP cell during the cycling test is also illustrated in **Figure 6d**: a promising average coulombic efficiency of 99.95% was observed, demonstrating the high stability of the LiBFSIE polymer gel electrolyte operated up to 4.0 V for 200 cycles. In comparison, the Li/LFP cell equipped with Gen2 electrolyte (**Figure S11**) showed a capacity retention of 78% after 170 cycles, with an average coulombic efficiency of 98.68%. This relatively low coulombic efficiency could be associated with unstable SEI, which continuously breaks down and reforms during cycling, on the Li metal. It is worth mentioning that this *in situ* LiBFSIE-electrode preparation technique can also be applied in making pouch cells. **Figure 6e** demonstrates the fine LiBFSIE-coated LFP laminate and Li metal electrode. No separators or liquid electrolytes were required in fabricating pouch cells. After being charged to 4

V, the LiBFSIE-based pouch cell successfully powered a commercial light wire (**Figure 6f, Video S3**). The cycling performance of the pouch cell is illustrated in **Figure S12**.

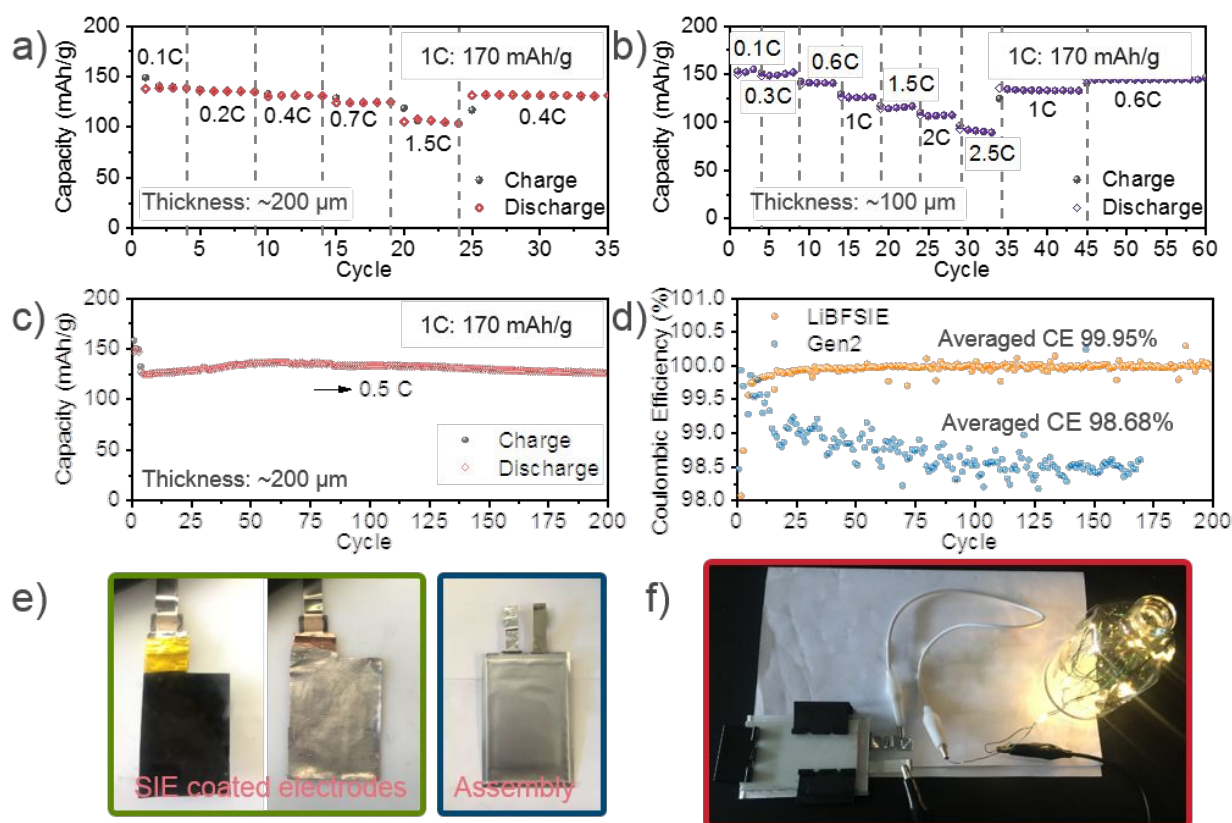


Figure 6. Battery performance of Li/LiBFSIE/LFP cells. a, b) Rate capability tests of Li/LiBFSIE/LFP cells at various current densities. A better rate capability is observed when a thinner LiBFSIE coating is applied. 1 C corresponds to 170 mAh/g. c) Cycling performance of the Li/LiBFSIE/LFP cell at 0.5 C. d) Coulombic efficiency of a long-cycling Li/LiBFSIE/LFP cell (orange) and a Li/Gen2/LFP cell (blue). e) LiBFSIE coating on electrodes for pouch cells. f) Decorative lights powered by the Li/LiBFSIE/LFP pouch cell.

It is known that the SEI formation is critical to the stability of the Li metal electrode and to battery performance with respect to coulombic efficiency, safety issues, etc.^{59,60} To further investigate the reductive decomposition of the LiBFSIE gel electrolyte with respect to the Li metal electrode

during cycling, X-ray photoelectron spectroscopy (XPS) characterizations were carried out on the Li metal electrodes of the Li/LiBFSIE/LFP cell and the Li/Gen2/LFP cell after 20 cycles. A freestanding LiBFSIE film was employed as the electrolyte so that it could be entirely peeled off from the Li metal electrode for the XPS study. Assignments of peaks with binding energy are according to the NIST XPS database and other published work.^{61,62} In the fluorine 1s spectra (**Figure 7a and 7b**), the peaks located at 685 eV are assigned to the F-Li bond from LiF. An obvious peak at 687 eV observed in the LiBFSIE cell can be associated with the organic fluorine component.⁶³ Considering the LOMO levels of FEC and LiBFMB in LiBFSIE, both of them would go through a reduction on the Li metal surface. A control experiment with Gen2 (10 wt% FEC) was also carried out (**Figure S13**). Similarly to Gen2 electrolyte, only LiF was observed and the peak at 687 eV disappeared in the absence of LiBFSIE, revealing that the organic fluorine component resulted from the decomposition of LiBFMB. This newly generated organic fluorine is believed to facilitate uniform Li deposition and promote coulombic efficiency.^{64,65} In the carbon 1s and oxygen 1s spectra (**Figure 7b,c,e, and f**), a higher carbonyl-group content was observed with respect to the ether on the Li metal electrode cycled in Gen2 than that cycled with LiBFSIE, indicating the further decomposition of the organic solvent, which may form thicker SEI layers.

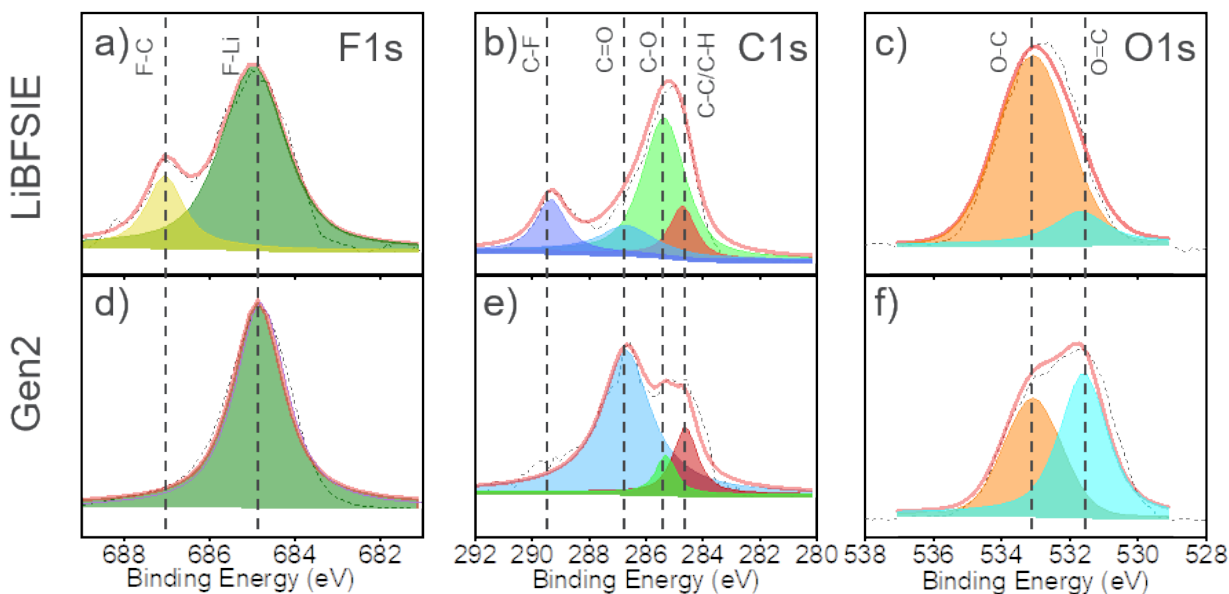


Figure 7. XPS characterizations of cycled Li metal electrodes in Li/LiBFSIE/LFP cell and Li/Gen2/LFP cell. a-c) F1s, C1s and O1s spectra of cycled Li metal electrode in Li/LiBFSIE/LFP cell. d-f) F1s, C1s and O1s spectra of cycled Li metal electrode in Li/Gen2/LFP cell. For F1s spectra, the green peak corresponds to LiF (F-Li, 685 eV) and the yellow peak is assigned to organic fluorine (F-C, 687 eV). The C1s spectra are deconvoluted into a red peak corresponding to hydrocarbon (C-H/C-C, 284.8 eV), a light green peak corresponding to ether carbon (C-O-C, 285.8 eV), a blue peak corresponding to a carbonyl group (C=O, 286.9 eV), and a purple peak corresponding to the C-F bond at 298.8 eV. For O1s spectra, the orange peak at 533.2 eV is attributed to the O-C bond and the light blue peak corresponds to the O=C bond at 531.4 eV.

The outstanding electrochemical performance of the LiBFSIE polymer gel electrolyte in conjunction with a Li metal anode and a LFP cathode can be attributed to the following factors: From the polymer gel electrolyte design perspective, the low dissociation energy between boron and Li promoted ion solvation by the ester groups and plasticizers, and contributed to the high ionic conductivity of the polymer gel electrolyte. Meanwhile, the fluorinated Li salt monomer

enabled the lower HOMO level of the polymer gel electrolyte, improving the electrochemical stability on the cathode; and led to the formation of organic fluorine components in the SEI to protect the Li metal electrode. The immobilization of anions on the polymer backbone endowed the polymer gel electrolyte with a high t_{Li^+} and avoided the side reactions induced by the concentration gradients. Meanwhile, the uniformity of the Li deposition was enhanced by the homogeneous Li ion distribution in this SIC. From the engineering perspective, the *in situ* integration of polymer gel electrolyte with electrode materials improved the interface contact, facilitating the fast Li ion exchange.

CONCLUSION

In this work, a boron- centered fluorinated SIC polymer gel electrolyte was rationally designed to achieve high t_{Li^+} and electrochemical stability. The as-prepared LiBFSIE achieved an ionic conductivity of 2×10^{-4} S/cm at 35°C, which is mostly contributed by Li ions owing to the high t_{Li^+} of 0.93. The unique chemistry involved in preparing LiBFSIE also enabled the superior performance of the cross-linked SIC electrolyte in comparison to the non-cross-linked binary-ion systems. According to the simulation and experimental results, an obvious enhancement of ion diffusion was observed and the concentration gradient was mitigated when LiBFSIE was applied. The interfacial properties were optimized with an *in situ* UV-vis-initiated polymerization, together with electrode materials that enabled wetting and infiltration. A stable Li stripping/plating test also demonstrated the uniform Li deposition assisted by LiBFSIE. By virtue of the high t_{Li^+} and electrochemical stability (up to 5 V), a promising average coulombic efficiency of 99.95% was achieved in the Li/LiBFSIE/LFP cell for 200 cycles; this efficiency is much higher than that of a cell with liquid electrolyte (Gen2). This observation can be attributed to the organic fluorine

components in the SEI between Li metal and LiBFSIE, which are favorable for Li nucleation and plating. The method of design and preparation of highly conductive and stable LiBFSIE could also be expanded to design novel polymer electrolytes, polymer binders, and artificial SEIs for next-generation batteries.

EXPERIMENTAL SECTION

Materials

Allyl bromide, methyl tert-butyl ether (MTBE), sodium chloride, sodium hydroxide, and lithium carbonate were purchased from Sigma Aldrich and used without purification. Ethanol, boric acid, and diethyl carbonate (DEC) from Alfa-Aesar, 3-Å molecular sieves (8–12 mesh) from JT Baker, Gen2 (1.2 M LiPF₆ in EC/EMC(3:7)) from Tomiyama Pure Chemical Industries, Japan, and diethyl 2-fluoromalonate from Synnovator were used as received.

Synthesis of LiBFMB

A 500-mL, 3-neck round-bottom flask was charged with a solution of sodium ethoxide (126 g, 20% in ethanol), followed by ethanol (150 ml). Diethyl 2-fluoromalonate (60 g) was added over 5 min, with a 1–2°C increase in temperature. The mixture was stirred for 1 h at room temperature, forming a precipitate. Allyl bromide (32 ml) was added via a syringe pump over 1 h, using a room-temperature water bath to moderate the exotherm. After the addition was completed, the reaction mixture was heated to 55°C for 1 hour, then cooled overnight. The ethanol was evaporated, and the residue was partitioned between MTBE and water. After the MTBE was removed, the product was distilled at 77°C, 5 torr pressure to give 65 g of a clear, colorless liquid (Product 1 in **Figure S4**). A solution of sodium hydroxide (26.78 g, 0.61 mol, 2.22 equiv) in water (150 ml) was prepared,

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3 stirring until all solids were gone. Ethanol (450 ml) was added, followed by diethyl 2-allyl-2-
4 fluoromalonate (64.6g). The mixture was stirred for 1 h at 60°C, then cooled to <15°C and filtered.
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6 The solids were washed with ethanol (2 x 150 ml) and MTBE (2 x 100 ml), then dried in a vacuum
7 oven to yield 60.61 g (Product 2 in **Figure S4**). 2-Allyl-2-fluoro malonic acid, disodium salt (60 g)
8 was slurried in water (60 ml) and just enough HCl (12M, 52 ml, 2.22 equiv) to make a thick paste.
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10 The remainder of the HCl was then added dropwise until a clear colorless solution (pH = 1) was
11 obtained. The mixture was stirred for 1 h, then saturated with solid sodium chloride. The aqueous
12 solution was extracted with MTBE (3 x 100 ml), dried with sodium sulfate, and evaporated by
13 rotary evaporation. The residual solids were washed with dichloromethane and dried to yield 41.48
14 g (Product 3 in **Figure S4**). A 500-ml, 2-necked round-bottom flask equipped with a thermocouple
15 probe was charged with a thoroughly ground mixture of 2-allyl-2-fluoro malonic acid (10.0009 g,
16 61.68 mmol, 1.00 equivalent), Li carbonate (1.143 g, 15.47 mmol, 0.25 eq), and boric acid (1.9085
17 g, 30.87 mmol, 0.50 eq). The solids were suspended in DEC (250 ml) with stirring. A Dean-Stark
18 trap filled with 3-Å molecular sieves was attached to the reaction flask and the mixture heated to
19 reflux under reduced pressure for 6 h. The reaction mixture was then cooled under nitrogen and
20 stirred overnight. The mixture was filtered, washed with 20 ml of fresh DEC, and then again with
21 20 ml dry MTBE before drying in a vacuum oven at 60°C; 8.3 g of white powder was obtained
22 (Product 4 in **Figure S4**). To purify LiBFMB salt, 3 g crude material was added to 50 ml near-
23 boiling acetonitrile (ACN) at 80°C and stirred for 30 min. The suspension was then filtered while
24 hot, and the filtrate was collected and concentrated with rotary evaporation; 1.7 g LiBFMB salt
25 was obtained after further drying under vacuum for 12 h.
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54 **Method to Prepare Cross-linked Polymer Gel Electrolyte**

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A 2-ml mixture of PC, EC, and FEC with a volume ratio of 6:3:1 was added to a mixture of 2 mmol LiFBMB, 1.32 mmol TMPT, and 0.2 mmol 2,2-dimethoxy-2-phenylacetophenone (DMPA) in an argon-filled glovebox and stirred for 30 min to achieve a clean solution precursor. Then the solution precursor was placed in the wells of customized Teflon molds and exposed to UV-vis light for 15 min. A freestanding, cross-linked polymer film was obtained.

Method to Prepare Cells with *In Situ*-formed LiBFSIE/Electrode Composite

For the coin cell fabrication, 10 μ l and 6 μ l of liquid precursor were drop-casted onto the Li metal (10 mm) and LFP electrode (5/16 in.) surface, respectively, via a fine pipette. A 15-min rest allowed liquid wetting and infiltration inside the electrode prior to the polymerization. After 15 min of UV-vis exposure, *in situ*-formed LiBFSIE/electrode composites were achieved. The total thickness of LiBFSIE was measured to be $\sim$ 200 μ m with a digital thickness gauge. (For $\sim$ 100- μ m LiBFSIE, 5 μ l and 3 μ l of liquid precursor, respectively, were applied to Li metal and LFP electrodes.) For pouch-cell fabrication, 250 μ l of liquid precursor was applied to the electrodes.

Li Metal Cell Fabrication.

LFP laminates were prepared by casting a slurry (LFP [80 wt%], C45 [10 wt%], and PVDF [10 wt%] in N-Methyl-2-pyrrolidone (NMP) onto aluminum foil with a doctor blade and drying at 80°C overnight. The solvent residue was further removed by heating the laminates at 110°C under vacuum for 12 h. The active-material mass loading is $\sim$ 2.4 mg/cm². Cells were then assembled by simply compressing the LiBFSIE/LFP and LiBFSIE/Li electrodes together without any separators. 2032-type SS coin cells were assembled in an argon-filled glovebox. For pouch-cell fabrication, the laminate was cut into a size of 3 cm $\times$ 4 cm as the cathode. A Li metal film was compressed

onto a copper foil with a size of 3.2 cm × 4.2 cm to form the anode. Nickel tabs and aluminum tabs were welded on as current collectors for the anode and cathode, respectively. The pouch cells were eventually sealed in a dry room.

Electrochemical Characterization

Electrochemical studies were performed on a Solartron Analytical 1400 Cell Test System. The polymer electrolyte was trimmed into a circular disk with a diameter of 16 mm, and the thickness of the disk was measured using a Digimatic indicator. A symmetrical SS/electrolyte/SS cell was then assembled in an argon-filled glovebox for EIS study. The impedance of the cell was measured in the frequency range of 1 MHz to 0.1 Hz at the open-circuit potential at an amplitude of 10 mV. The data generated from the measurements were then plotted in Nyquist plots. The ohmic resistance (R_{Ω}) of the sample, obtained from the intercept of the semicircle with the Z' axis at the low-frequency end of the Nyquist plot, was used to calculate the ionic conductivity via the following equation:

$$\sigma = \frac{L}{A} \times \frac{1}{R_{\Omega}} \quad ,$$

where L is the thickness and A is the cross-sectional area of the electrolyte disk.

The measurements were performed within the temperature range from 20 to 80°C at 20°C intervals. At each temperature, the sample was equilibrated for 15 min before measurement. The t_{Li+} of the gel conductor was calculated using a combination of DC polarization measurement and EIS. In the argon-filled glovebox, a symmetrical Li metal/gel disk/Li metal cell was assembled and was subjected to a constant polarization bias (ΔV) to determine the initial (I_0) and steady-state (I_{ss}) currents, while at the same time EIS was performed by applying a constant voltage between 1 MHz

and 0.1 Hz under open-circuit conditions to obtain the resistance of the gel disk and interface between the film and electrode before (R_0) and after (R_{ss}) polarization. Then t_{Li^+} was calculated using the following equation:

$$t_{Li^+} = \frac{I_{ss}(\Delta V - I_0 R_0)}{I_0 (\Delta V - I_{ss} R_{ss})}$$

FTIR, SEM, XPS, XPCS, and Mechanical Strength Characterization

FTIR spectra were collected on a Thermo Scientific Nicolet iS5 spectrometer placed in an argon-filled glovebox, using the attenuated total reflection model. The morphological characterizations were performed on a JEOL JCM-6000Plus scanning electron microscope. XPS characterizations were carried out by using a PHI 5000 VersaProbe II System (Physical Electronics) with a base pressure of 2×10^{-9} Torr. The Li metal electrodes were harvested after 20 cycles. Note that LiBFSIE was not coated *in situ* on the Li metal electrode for the XPS analysis, allowing the Li to be peeled off after cycling. The photoelectron spectra were achieved in the fixed analyzer transmission mode using Al K α radiation ($h\nu = 1486.6$ eV, 100- μ m beam, 25 W). Peak fitting was performed using Shirley background correction and the Gaussian–Lorentzian algorithm. XPCS was used to investigate the activation energy of the LiBFSIE electrolyte. A LiBFSIE film with a thickness of ~ 400 μ m was sealed in a Kapton pocket under argon and mounted on a heating stage at beamline 8-ID-I at the Advanced Photon Source, Argonne National Laboratory, for study over a temperature range from 25°C to 55°C. The x-ray energy was set at 11 keV. The activation energy was measured from the autocorrelation function (g_2) of speckles measured in the low x-ray momentum transfer limit ($q = 0.0033$ Å $^{-1}$), with a scattering contrast of 1.08. Mechanical strength tests were carried out on an Instron 3343 universal tester. LiBFSIE was cut into a film with a size

of 2 cm (l) $\times$ 0.5 cm (w) $\times$ 0.5 mm (d) and mounted onto the holder for a stretching test. For the adhesive study, two separate samples were attached. The overlapping area was half the size of each of the films. Stretching tests were conducted at a rate of 5 mm/min.

DFT Calculations

The molecular geometry was optimized at the DFT level with the B3LYP/6-311G** method using the Gaussian 09 program.⁶⁶ The energies of the HOMO and LUMO were obtained from the optimized geometry. The dissociation energy is defined by $E_d = E_{Li} + E_{anion} - E_{Li + anion}$; E_{Li} , E_{anion} , and $E_{Li + anion}$ are the energy of the Li⁺ ion, anion, and ion pair, respectively. The delocalized molecular orbitals HOMO and LUMO were visualized using the VMD 1.9.3 software.⁶⁷

Molecular Dynamics (MD) Methods

Figure S3a shows a schematic of the MD system, which features one BFMB⁻ ion and one Li⁺ ion immersed in PC. The simulation box is larger than 5 nm in all directions, which is large enough to obtain a bulklike behavior at the position away from the BFMB⁻ ion. The umbrella sampling method⁶⁸ was used to calculate the PMF as a function of the distance between the Li⁺ ion and the boron atom in the BFMB⁻ ion. The optimized geometry of the BFMB⁻ ion, symmetric in the *x*-direction, was fixed in the center of the simulation box. A series of configurations with different radial Li-B separations in the *yz*-plane were generated. A harmonic potential (force constant: 2000 kJ/(mol $\cdot$ nm²)) was applied to the Li⁺ ion to constrain its radial distance from the boron atom in the *yz*-plane. The position of the Li⁺ ion was recorded during a 5-ns simulation after a 5-ns equilibrium simulation for each configuration under constraint at 298 K. Finally, the PMF of the

Li⁺ ion was extracted from the histograms of its position in the different runs using the Weighted Histogram Analysis Method implemented in the Gromacs code.⁶⁹ The building of the force field for the Li⁺ ion, BFMB⁻ ion, and PC molecule was based on the OPLS-aa force field.⁷⁰ The partial charge distribution for the BFMB⁻ ion was computed by fitting the molecular electrostatic potential using a grid-based method at the DFT level with the B3LYP/6-311G** method using the Gaussian 09 program.⁶⁶

ASSOCIATED CONTENT

Supporting Information

The Supporting Information includes 13 figures, 4 tables, and 3 videos and can be found with this article online.

Note

The authors declare a provisional patent application (USPTO 16/567,647) owned by Argonne National Laboratory that discloses the LiBFSIE with a high t_{Li^+} and high electrochemical stability.

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